FUNCTORIALITY OF THE EQUIVALENCE BETWEEN QUASI-COHERENT MODULES ON A SCHEME AND ON ITS BIG AND SMALL SITES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Part (2) is a negative statement and its proof is an explicit counterexample; the counterexample lies inside the counted span and is reproduced verbatim, and it is recorded as a load-bearing obligation rather than as a remark.

Editorial note. Part (3) is proved in the statement only for the etale topology, the author noting that this is the case treated; the delivered text is reproduced as it stands and no case was added.

Editorial note. The statement displays a commutative diagram of categories of quasi-coherent modules; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The equivalences between quasi-coherent modules on S, on the big site and on the small site. Same chapter, Proposition proposition-equivalence-quasi-coherent, invoked in the statement at lines 3112-3118. Supplies the functors whose compatibility with pullback and pushforward is being asserted.
- Changing the topology on a ringed site. Same chapter, Remark remark-change-topologies-ringed-sites, invoked at proof lines 3138-3139. Gives part (1), the compatibility of the equivalences with pullback.
- Pushforward of quasi-coherent modules along a quasi-compact quasi-separated morphism. Schemes, Lemma schemes-lemma-push-forward-quasi-coherent, invoked at proof lines 3153-3156. Ensures that f and every base change of f preserve quasi-coherence, which is needed before the square in part (3) can even be formed.
- Flat base change for cohomology of coherent sheaves. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology, invoked at proof lines 3160-3161. Identifies the sections of the two sides of the asserted equality over an etale morphism into S.

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Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03DX: sources/stacks-descent/pinned-original-source.tex, source lines 2716–2716. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 03FH: sources/stacks-descent/pinned-original-source.tex, source lines 2587–2587. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Proposition 0.1. *Let $f : T \to S$ be a morphism of schemes.*

1. *The equivalences of categories of Proposition 03DX are compatible with pullback. More precisely, we have $f^*(\mathcal{G}^a) = (f^*\mathcal{G})^a$ for any quasi-coherent sheaf $\mathcal{G}$ on S.*
2. *The equivalences of categories of Proposition 03DX part (1) are* **not** *compatible with pushforward in general.*
3. *If f is quasi-compact and quasi-separated, and $\tau \in \{Zariski, étale\}$ then f_* and $f_{small,*}$ preserve quasi-coherent sheaves and the diagram*

$$\begin{array}{ccc} QCoh(\mathcal{O}_T) & \xrightarrow{f_*} & QCoh(\mathcal{O}_S) \\ {\scriptstyle \mathcal{F} \mapsto \mathcal{F}^a}\downarrow & & \downarrow{\scriptstyle \mathcal{G} \mapsto \mathcal{G}^a} \\ QCoh(T_\tau, \mathcal{O}) & \xrightarrow{f_{small,*}} & QCoh(S_\tau, \mathcal{O}) \end{array}$$

is commutative, i.e., $f_{small,*}(\mathcal{F}^a) = (f_*\mathcal{F})^a$.

Proof. Part (1) follows from the discussion in Remark 03FH. Part (2) is just a warning, and can be explained in the following way: First the statement cannot be made precise since f_* does not transform quasi-coherent sheaves into quasi-coherent sheaves in general. Even if this is the case for f (and any base change of f), then the compatibility over the big sites would mean that formation of $f_*\mathcal{F}$ commutes with any base change, which does not hold in general. An explicit example is the quasi-compact open immersion $j : X = \mathbf{A}^2_k \setminus \{0\} \to \mathbf{A}^2_k = Y$ where k is a field. We have $j_*\mathcal{O}_X = \mathcal{O}_Y$ but after base change to $\mathrm{Spec}(k)$ by the 0 map we see that the pushforward is zero.

Let us prove (3) in case $\tau = étale$. Note that f, and any base change of f, transforms quasi-coherent sheaves into quasi-coherent sheaves, see Schemes, Lemma 01LC. The equality $f_{small,*}(\mathcal{F}^a) = (f_*\mathcal{F})^a$ means that for any étale morphism $g : U \to S$ we have $\Gamma(U, g^*f_*\mathcal{F}) = \Gamma(U \times_S T, (g')^*\mathcal{F})$ where $g' : U \times_S T \to T$ is the projection. This is true by Cohomology of Schemes, Lemma 02KH. □